

Exploring the Gap between Real and Virtual Nature

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ABSTRACT

We describe the design and preliminary findings of an experiment that aims to elucidate the cost of omitting accurate haptic and olfactory stimulation from VR nature immersion experiences. Across four separate sessions, participants are immersed in virtual urban and forest environments, while seated both outside in real nature and indoors, with the correspondence of virtual environment to real environment and the order of presentation counterbalanced. We compare multiple restorative outcome measures between the four conditions, including physiological data (EDA and HR), subjective surveys of stress and contentment, and objective performance on tests of visual and auditory attentional resources.

Keywords: Virtual Reality, Forest Bathing, Nature Immersion.

Index Terms: Human-centered computing – Human computer interaction (HCI) – Interaction paradigms – Virtual reality.

1 INTRODUCTION

International research has shown multiple benefits to health and well-being from exposure to natural environments. Shinrin-yoku, the term coined by the Japanese Ministry of Agriculture, Forestry, and Fisheries in 1982, is defined as taking in the forest atmosphere or forest-bathing [6]. VR-based nature exposure has the potential to provide value for people with limited access to nature, such as urban residents, the elderly, and those who are incarcerated.

Our aim in assessing the relative role that non-visual and non-auditory stimuli play in the restoration process, is to help gain insight into why virtual reality nature immersion has been found less effective than real world nature immersion, and to elucidate the extent to which our current inability to replicate more than the visual and auditory sensations of being outdoors represents a fundamental limitation in virtual reality's potential. Specifically, this experiment allows us to establish an upper bound on the additional benefit that could potentially accrue from providing perfectly realistic haptic and olfactory input, along with the unambiguous certainty that one is actually physically outdoors.

2 RELATED WORK

Previous research has substantiated the restorative value derived from exposure to natural environments. Even short-term exposure has been found to result in multiple positive health outcomes. Virtual reality-based nature exposure offers an alternative route to obtain similar benefits, although not to the same extent [1].

Still, over 100 experiments working with nature-based pictures, videos, or immersive virtual environments report a connection to enhanced mood, executive cognitive functions, decreased stress, or reduced pain with minimal or no side effects [1].

3 METHODS

Our hypotheses are as follows: All of the restorative outcome measures will be “better” in the virtual nature immersion condition (VN)

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(a) Virtual Nature Environment



(b) Virtual Urban Environment

Figure 1: Virtual Environments



(a) Physical Outside Environment



(b) Physical Inside Environment

Figure 2: Physical Environments

than in the virtual urban immersion condition (VU), both when experienced indoors (PI) and when experienced outdoors (PO), where “better” is defined according to each measure. Considering the virtual nature and virtual urban environments separately, all of the restorative outcome measures will be “better” when the experience takes place outdoors (PO) vs indoors (PI). Overall, the restorative outcome measures will be highest in the outdoor/virtual nature condition combination (VN/PO) and lowest in the indoor/virtual urban condition (VU/PI). Between the VN/PI and VU/PO conditions, restorativeness outcomes will be dominated by the visual and auditory input, so that the outcome measures are higher in VN/PI than in VU/PO. Measures of presence will be highest when the virtual nature environment is experienced while physically outdoors (VN/PO); no other between-condition differences are expected. The outcome measures of greatest interest are:

1. How much is gained by experiencing the virtual nature environment while outside in real nature compared to experiencing the virtual nature environment while inside?
2. What is the gain, or is there a gain, when experiencing the virtual urban environment while outside in real nature compared to experiencing the virtual urban environment while inside?

3.1 Materials

The experiment uses the same HP OMEN X Compact Desktop PC, Meta Quest 2 HMD, Empatica E4 wristband, and Bose QuietComfort Noise-cancelling headphones in the outdoor and indoor environments. The perceived restorativeness questionnaire is administered in the virtual environment, while the other surveys and the objective performance tasks are administered outside of VR on a Dell XPS laptop. The stressor task uses the laptop to send audio to the Bose headphones and the HMD for the visual component.

3.2 Environments

Virtual environments on the Unreal Engine Marketplace were chosen to represent the nature (figure 1a [7]) and urban (figure 1b [2]) environments after testing for similar performance and verifying that each offers a similar number of interesting things to look at.

The experiment took place at two locations on campus. For the outdoor environment, shown in figure 2a, we identified an area of a public park adjacent to the University of Minnesota campus that was mostly devoid of people. It was adjacent to a river and separated from vehicular traffic and buildings. For the indoor environment, shown in figure 2b, we utilized a quiet computer lab nearby.

3.3 Procedure

The study design was within-subjects, and the order of the 4 conditions was counter-balanced among participants. Participants completed 2 conditions on 2 different days, separated by at least 48 hours. Each day involved both the outdoor and indoor locales, and both the virtual nature and virtual urban scenes, but in different combinations. This was done to try to blind participants to the hypotheses of the experiment, which is important when collecting subjective responses. Before the first session, the participant was sent the informed consent form to review. When participants arrived, the experimenter went over the consent form and obtained the participant's signature. They were reminded that they could drop out of the experiment at any time. The Institutional Review Board approved this experiment.

Each of the 4 blocks went as follows, with the exception of the task explanation only happening in the first block. Baseline measures of heart rate and galvanic skin conductance (tonic EDA) were recorded. The participants then indicated their subjective stress and contentment levels on a 10-point scale. After an introduction to the objective performance tasks and the stressor task, participants completed the stressor task and then updated their stress and contentment levels. This was followed by the objective performance tasks. Next, participants spent 5 minutes immersed in the virtual environment, while seated in a comfortable camp chair with their shoes off to make contact with the ground. Finally, they completed the perceived restorativeness survey in the virtual environment, then took off the HMD, repeated the objective performance tasks, and answered the stress and contentment questions, as well as a presence survey. If this was the first block of the day, participants would then travel on foot to the second location, with the additional goal to clear their mind. After the fourth block, participants completed an exit questionnaire asking for demographic information and an assessment of preference and overall satisfaction with the experience.

3.3.1 Stressor Task

This task was based on the Trier Social Stress Test [5] and used to increase stress levels before the virtual immersion. Participants began by preparing for the first part of a 2-part mock job interview. Each time, they were given one of the 4 questions below and asked to spend 1 minute mentally preparing an answer. The interview questions, all of which are eventually used in randomized order, are:

- Why should we hire you?
- What is your greatest weakness?
- Can you tell us about a time when you failed at an important task and why?
- What is the most critical feedback you received from a boss and how did you respond to it?

After the 1 minute of preparation, the participant puts on the VR headset and sees a virtual job interview scenario, facilitated by the VirtualSpeech app [10]. The participant spends 2 minutes giving their answer to the virtual avatars, who do not speak. In the second part, the participant then spends 3 minutes counting backward by a given number, starting at a given number, while hearing distracting noises of people talking, traffic, and building construction. The pairings of numbers to count backwards from and by respectively were: 1022/13, 1021/14, 1023/14, 1011/13. The experimenter presses a key to play a buzzer sound if the answer is wrong or if the participant fails to answer within 7 seconds. Then the participant starts over. To increase stress, the experimenter would occasionally invoke the buzzer when the answer was correct and on time.

3.3.2 Outcome Measures

The Perceived Restorativeness Scale (PRS) [4] was administered within the virtual environment after 5 minutes of immersion to assess the restorative impact of the VR experience. The first objective performance task was a visual version of Kirchner's "n-back" task [3] which measures the capacity of an individual's working memory and their ability to manipulate information stored in it; we used $n = 2$. The second objective performance task was an auditory backward digit span (BDS) test, starting at $n = 2$ and increasing in length until the participant failed two consecutive trials [8]. EDA and HR were gathered continuously throughout the experiment using an Empatica E4 wristband with custom finger-mounted sensors, with the following key time points marked: the end of the baseline period, the end of the stressor task, and the end of VR immersion.

3.3.3 Presence Measure

After the PRS and attentional tasks, participants completed a presence questionnaire comprised of a subset of the iGroup presence questionnaire (IPQ) [9] to measure the sense of presence experienced in a virtual environment.

4 PRELIMINARY RESULTS

We were only able to collect data from 4 participants before the outdoor weather became seasonally unpleasant; we plan to resume data collection when warm weather returns in June. In the initial data, our results suggest a stronger effect of the visual/auditory stimuli than of the physical setting in which they are experienced, but also some possible additional benefits from actually being outdoors.

5 DISCUSSION AND CONCLUSIONS

Virtual forest bathing has the potential to offer some of the benefits of real nature immersion in situations where the latter is impossible, such as during chemotherapy, hemodialysis, incarceration, etc. A better understanding of the fundamental capabilities and limitations of present VR technologies can help to inform the most urgent directions for future efforts to achieve improved outcomes.

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